INCH-POUND

MIL-B-49436/6C(ER)
10 December 1993
SUPERSEDING
MIL-B-49436/6B(ER)
12 September 1991

MILITARY SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED NICKEL CADMIUM, BB-516/U

This specification is approved for use by Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49436

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

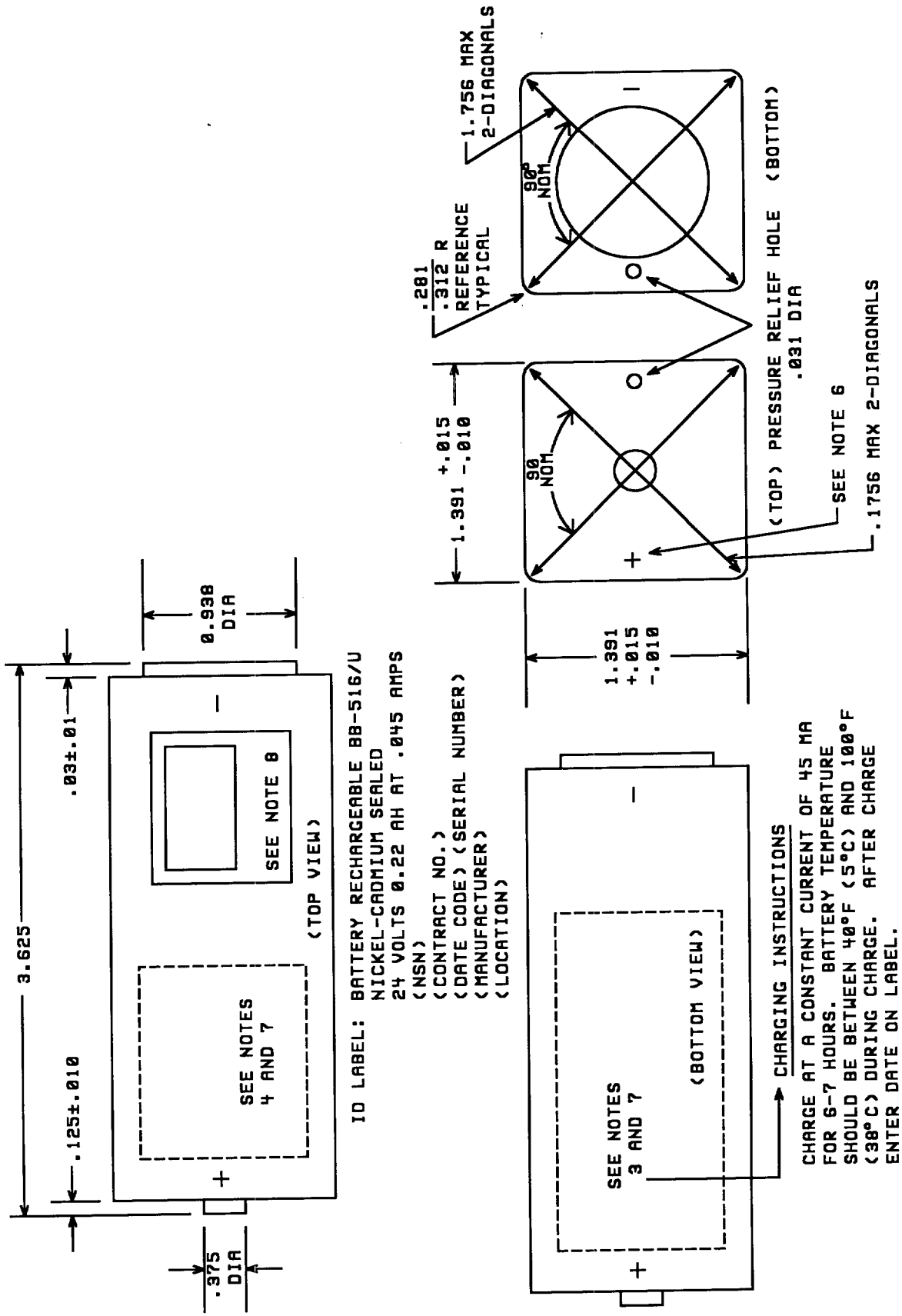


FIGURE 1. BATTERY DIMENSIONS AND LABELS.

Source: https://assist.dla.mil - Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Rated capacity (C/5): 0.22 AH.
Voltage (nominal): 24 volts.
Weight (maximum): 0.27 kg (0.59 lbs).
Terminals: 2 each MIL-N-46025 nickel, flat surface.
Number and type of cells: 20 sealed "1/3 A" size cells or equal in series.
Charge rate for testing and on battery (C/5 for 6-7 hrs): 0.045 amps.

Discharge Rates	5 Second Voltage(V)	Final Voltage(V)	Current (Amps)
C/5	-	20.0	0.045
C_5	-	20.0	0.20
$4C_5$	22.0	20.0	0.80
$8C_5$	-	20.0	1.60

Test	Discharge Rate	Capacity Requirement (AH)
Full Capacity Discharge	C_5	0.20
Cycle Life, 224 cycles	C_5	0.15
Overcharge, 24 hours	C_5	0.20
High Rate Discharge	$4C_5$	0.13
Low Temperature Discharge at -30°C (-22°F)	C_5	0.07
Retention of Charge 7 days at 50°C (122°F)	C_5	0.12
Pulse Discharge 1.6 amps, 5 seconds on, 25 seconds off, continuously to 20 volts	$8C_5$	0.13
Vibration	C/5	0.22

Mounting screws for connector: Not applicable.

Terminal markings: The positive terminal marking may be painted red. The negative terminal may be painted black.

Terminal: The two terminals shall be low resistance flat contacts of nickel per MIL-N-46025.

Electrical leakage: Not applicable.

Immersion: Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.

DRAWING AND SPECIFICATION NOTES

1. Except as noted, all values are minimum.

2. All charges and discharges shall be performed on fully assembled batteries through the terminals.

3. Charging instructions shall be provided on a label which will be attached to side of the battery opposite ID label.

4. Identification label shall include complete label information required in specification. It shall be located as shown on drawing and surface shall be designated as top of battery. If necessary, adjacent side may contain continuation of label content.

5. Dimensions are in inches and unless otherwise specified decimal tolerances are ± 0.031 in.

6. Symbol + on this surface; - on opposite end.

7. Labels may be an integral part of the container provided marking is permanent and clear or labels shall be separate, made of KOH resistant, non-metallic, material with permanent and clear marking.

8. Writeable charge history label.

9. The diagonals for top and bottom views are 1.756 inches maximum. Diagonals shall not exceed 1.756 inch maximum along the length of battery before or after testing or discharge.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL:

Custodian:
Army - ER

Preparing Activity:
Army - ER

Review Activities:
Army - CR, AV, GL

(Project 6140-A786)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:06Z
Check the source to verify that this is the current version before use.